

ANN ARBOR, MI, USA

WMO: 725374

Lat: 42.223N

Lon: 83.744W

Elev: 839

StdP: 14.26

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 94889

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.1	(c) 3.6	(d) -8.9	(e) 3.5	(f) -1.0	(g) -2.8	(h) 4.9	(i) 6.2	(j) 25.1	(k) 24.4	(l) 22.8	(m) 24.2	(n) 4.7	(o) 230	(p) 0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.6	(c) 89.8	(d) 73.4	(e) 87.1	(f) 72.5	(g) 84.0	(h) 70.7	(i) 76.4	(j) 85.2	(k) 74.7	(l) 82.9	(m) 73.0	(n) 81.0	(o) 10.3	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	128.2	81.0	72.2	122.8	79.4	70.4	115.3	77.6	40.3	85.1	38.6	83.0	37.1	81.2	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	19.1	17.1	-12.0	93.1	6.6	3.0	-16.8	95.3	-20.7	97.1	-24.4	98.8	-29.2	101.0
			-12.3	79.5	6.8	1.8	-17.2	80.7	-21.1	81.8	-24.9	82.8	-29.9	84.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	23.5	25.1	35.3	47.7	58.3	66.9	70.8	69.3	62.3	50.4	39.2	29.1
	DBStd	18.89	11.49	10.93	10.68	8.78	8.18	6.34	5.79	5.16	7.58	8.69	9.08	10.03
	HDD50	3250	822	697	472	143	19	0	0	0	4	105	339	649
	HDD65	6654	1286	1116	921	521	239	53	15	18	139	461	773	1112
	CDD50	2630	1	1	16	75	277	507	646	598	374	116	16	3
	CDD65	557	0	0	0	3	32	110	196	151	58	7	0	0
	CDH74	5852	0	0	10	57	406	1177	2114	1432	582	74	0	0
	CDH80	1759	0	0	1	7	94	362	737	382	163	13	0	0
(14) Wind	WSAvg	7.6	8.8	8.8	8.7	9.0	7.4	6.6	6.0	5.8	5.9	7.3	8.0	8.5
(15) Precipitation	PrecAvg	35.0	2.2	1.9	2.8	3.6	3.4	3.5	3.3	3.4	3.1	2.4	2.8	2.7
	PrecMax	47.9	5.0	6.2	6.1	7.2	7.8	8.5	6.7	8.7	7.0	7.6	5.9	5.0
	PrecMin	19.8	0.2	0.2	0.9	0.5	1.1	0.6	0.4	0.3	0.3	0.1	0.7	0.4
	PrecStd	5.2	1.2	1.2	1.2	1.4	1.6	1.6	1.7	2.1	1.5	1.3	1.3	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.9	59.4	73.2	80.7	86.5	91.2	94.5	91.0	89.7	81.5	69.4	59.4
		MCWB	51.1	52.3	61.6	63.9	70.7	73.5	74.8	73.7	72.6	69.6	57.6	55.9
	2%	DB	48.9	51.5	63.9	74.3	82.5	88.0	90.2	87.5	84.7	75.0	63.1	53.2
		MCWB	46.2	48.0	53.9	59.6	67.6	72.5	74.2	73.0	70.7	63.7	54.8	49.7
	5%	DB	43.4	44.7	57.4	69.7	79.0	84.1	87.6	84.0	80.8	70.5	58.5	47.1
		MCWB	40.4	40.6	49.9	56.7	66.0	70.5	73.0	71.2	68.1	61.2	52.1	43.9
	10%	DB	38.5	39.1	52.3	64.2	74.2	81.0	84.1	81.6	76.7	65.9	54.1	42.7
		MCWB	36.2	35.5	46.4	53.7	63.4	68.7	71.1	69.9	66.2	58.7	48.9	39.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	54.4	62.4	66.3	73.5	76.6	79.0	78.0	75.8	70.9	60.8	56.5
		MCDB	54.6	56.6	72.5	76.6	81.7	87.0	88.9	86.7	85.4	79.6	65.6	59.2
	2%	WB	46.3	47.6	56.4	62.2	70.3	74.3	76.9	75.4	73.2	66.8	56.6	50.2
		MCDB	47.8	50.8	61.7	69.5	79.3	84.1	86.4	82.3	80.9	72.4	61.7	52.4
	5%	WB	40.6	40.7	51.7	58.8	67.8	72.6	75.1	73.5	70.7	62.9	52.8	44.4
		MCDB	43.2	44.2	56.0	67.5	76.3	81.8	83.3	80.7	77.0	67.8	57.1	47.0
	10%	WB	36.4	35.9	46.1	55.1	65.2	70.4	73.2	71.8	68.4	59.5	49.1	39.4
		MCDB	38.5	38.8	51.4	63.2	72.5	78.1	81.1	78.7	74.2	64.8	53.8	42.0
(35) Mean Daily Temperature Range	5% DB	MDBR	15.0	16.5	18.5	21.7	22.5	22.6	23.6	22.3	23.5	21.3	17.4	13.5
		MCDBR	17.4	21.2	27.1	30.6	27.9	26.8	26.3	25.3	27.8	27.2	24.3	18.2
	5% WB	MCWBR	15.2	17.4	18.7	17.9	15.2	13.5	12.2	12.4	14.5	16.6	18.0	15.6
		MCWBR	16.7	20.6	25.0	25.7	24.0	23.0	22.1	20.8	22.6	23.0	21.3	16.9
(40) Clear-Sky Solar Irradiance	taub		0.297	0.337	0.355	0.401	0.428	0.450	0.441	0.431	0.386	0.353	0.316	0.310
		taud	2.390	2.252	2.328	2.218	2.195	2.170	2.216	2.232	2.342	2.421	2.482	2.383
	Ebn at Noon		269	271	281	275	270	263	264	263	268	264	260	252
		Edh at Noon	25	34	36	43	45	47	44	42	35	29	23	23
(44) All-Sky Solar Radiation	RadAvg		487	767	1121	1460	1702	1914	1931	1671	1354	882	553	386
	RadStd		45	98	111	129	150	123	96	67	119	78	73	35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	N/A

Nomenclature: See separate page